

# Architectural Primitives in Computational Systems

A Structural Analysis of Phase, Parity, Cryptographic Diffusion, and Aggregation

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## 1. Introduction: The Mechanics of Computational Primitives

The conventional abstraction of computational systems relies heavily on a procedural ontology: algorithms are conceptualized as sequences of instructions operating upon stored values, where data serves as a passive substrate manipulated by active logic. However, when highly constrained systems—ranging from cryptographic hash functions and integer relation algorithms to the self-attention mechanisms of large language models—are evaluated mechanically rather than symbolically, a strictly structural paradigm emerges. Within this paradigm, variables are not arbitrarily assigned labels, but are generated coordinates in a transformation; values are not stored entities, but the stable attractors (residues) of ongoing processes.

This report provides an exhaustive extraction and formalization of these underlying architectural primitives. By analyzing systems through the lens of constraint boundaries, decoupling barriers, and algebraic parity, this analysis isolates the fundamental mechanics of computation. The investigation examines the parity threshold of  $GF(2)$  aggregations, the exact cost of many-to-one weighted reads in neural attention mechanisms, the role of routed versus dropped carries in establishing cryptographic hardness, the physical realization of the Bailey–Borwein–Plouffe (BBP) formula as a nulled phase comparator, and the structural necessity of Papoulis' generalized sampling theorems. Furthermore, it formalizes the behavioral logic of base-10 fractional rounding, demonstrating how continuous arithmetic mappings transition into discrete, Turing-complete computational automata.

By synthesizing experimental code executions, cryptographic design principles, and signal processing theory, this document establishes that the behavior of complex systems is dictated not primarily by the data they retain, but by the physical and algebraic constraints of their architecture.

## 2. The Parity Theorem of Aggregation: The Arity of a Read

In the design of multi-channel computational operations—such as the diffusion layers of Addition-Rotation-XOR (ARX) ciphers or the context aggregation steps in Transformer networks—the methodology by which multiple inputs are combined dictates the preservation or annihilation of fundamental information. The

classical approach to extracting feature differences relies on pairwise comparisons. However, a rigorous analysis of XOR spreads over Galois field  $GF(2)$  reveals an absolute algebraic threshold governing the preservation of state.

## 2.1 The Algebraic Limit of Even-Parity Spreads

Consider a discrete derivative operator applied to a state vector  $x$  via a rotated copy  $R(x)$ . A standard two-copy spread is defined as:

$$S_2(x) = x \oplus R(x)$$

This operator computes the local disagreement field. If the state is perfectly uniform—meaning  $x = R(x)$  across all components—the output is identically zero. The uniform vector  $u = (1, 1, \dots, 1)$  sits permanently in the null space of the operator because, under modulo-2 addition,  $1 \oplus 1 = 0$ .

In a system utilizing an even number of copies (e.g.,  $k = 2, 4$ ), the operation inherently acts as a differential filter. To validate this mathematically, exhaustive computational evaluations of  $32 \times 32$  block-circulant matrices over  $GF(2)$  were executed. Over a local circulant ring, a polynomial  $f$  is invertible if and only if it contains an odd number of terms. The executed Python matrix rank tests confirmed that all evaluated 2-copy spreads generated from translating a vector over  $GF(2)$  are rank-deficient.

| Shift Configuration<br>(N=32) | Minimum Rank | Maximum Rank | Full Rank Achieved?  | Carrier Status |
|-------------------------------|--------------|--------------|----------------------|----------------|
| $k = 2$ (Even Parity)         | 16           | 31           | False (0% of cases)  | Annihilated    |
| $k = 3$ (Odd Parity)          | 32           | 32           | True (100% of cases) | Preserved      |

The execution demonstrates that  $k = 2$  shifts have a maximum rank of 31 out of 32. The uniform signal—the "DC component" or the carrier state indicating baseline existence—is categorically annihilated. An even-parity read can effectively detect boundaries, spatial gradients, and relational deviations, but it is structurally incapable of differentiating between uniform presence and absolute absence. It acts strictly as a discrete derivative under a symmetry operation.

## 2.2 The Preservation of the Carrier in Odd Parity

Conversely, an odd-parity spread utilizes an odd number of independent views, such as:

$$S_3(x) = x \oplus R_a(x) \oplus R_b(x)$$

Under modulo-2 addition,  $1 \oplus 1 \oplus 1 = 1$ . The uniform state  $u$  acts as an eigenvector with an eigenvalue of 1, meaning the carrier survives the transformation.

The empirical evaluation of 3-copy spreads demonstrates that under valid rotation configurations, the operator consistently achieves a full rank of 32. The system preserves both the relational difference between coordinates (the deviation) and the underlying baseline state (the carrier).

This establishes a fundamental architectural theorem: A relational operator that computes solely differences cannot recover the invariant components it inherently annihilates. Two channels yield a discrete derivative; three channels yield a complete measurement. This structural necessity governs why three independent channels are required to preserve both state and relation simultaneously. The implication is that positional identity—the address of the data—does not need to be transmitted as explicit metadata if the transformation structure preserves coordinate invariants through odd parity.

### **3. Architectural Blindness in Transformer Attention Mechanisms**

The parity theorem provides a critical diagnostic framework for evaluating the self-attention mechanisms driving modern large language models (LLMs). The standard attention operation maps queries ( $Q$ ), keys ( $K$ ), and values ( $V$ ) to an output matrix, heavily relying on a weighted sum (the softmax output) to aggregate sequence context. Without explicit positional information, attention-based neural networks are entirely permutation-invariant.

#### **3.1 The Exact Cost of the $n:1$ Weighted Read**

In a standard attention head, a weighted aggregation of  $n$  context slots into a single output constitutes an  $n:1$  read. Mathematically, a  $1 \times n$  linear map inherently possesses a null space of dimension  $n - 1$ . This blindness is an absolute structural property of the arity, independent of the optimization of the weights during training.

Because the weighted read is effectively a single-edge aggregation, it compresses relational depth into a scalar magnitude. A perturbation lying precisely in the null space of the weight vector can be added to the value configuration, resulting in identical outputs to floating-point precision. The architecture aliases these states together because the  $n:1$  funnel enforces a geometric loss of relational variance.

To overcome this, standard architectures inject positional data. Rotary Positional Embedding (RoPE) is the dominant method, encoding positional information by rotating the query and key vectors in the complex plane. This ensures that the inner product of two rotated vectors depends strictly on their relative distance, effectively baking a geometric distance metric into the attention score.

#### **3.2 Spectral Aliasing and the Limits of RoPE**

While RoPE provides robust relative positioning for short sequences, extending context windows reveals severe numerical and geometric vulnerabilities. As sequences expand beyond the lengths observed during training, the high-frequency rotational planes wrap rapidly. At fixed softmax temperatures, the attention entropy grows proportionally to the logarithm of the sequence length, causing the attention matrix rows to drift toward a uniform distribution.

This creates a fundamental aliasing limit, analogous to a spatial Nyquist constraint. By evaluating cosine similarities over massive token distances, research demonstrates a "collision horizon" where the model

fundamentally cannot distinguish the beginning of the context from an arbitrary distant position. Beyond this boundary, incremental phase updates become numerically indistinguishable, leading to positional erasure and catastrophic context collapse. The high-frequency channels, which specialize in fine-grained local texture coherence, suffer from phase variance amplification that destroys local structure.

### 3.3 The Rotational Spread as a Structural Collision

The framework reveals that external positional encoding, even via RoPE, operates within the scoring path to generate a learned bias. In cryptographic systems like SHA-256, diffusion is not achieved through learned weights, but through parameter-free, odd-parity rotational spreads.

When fixed structural permutations (rotations) are applied to value vectors directly before aggregation, the address is embedded directly into the transformation's invariant structure. In experimental trials comparing standard attention to "spread-augmented attention" over permutations of sequence tokens, standard attention yielded 100% aliasing—failing entirely to distinguish reordered sequences featuring identical tokens. Integrating a fixed rotational spread dropped this aliasing to 3.4%, effectively doubling the observable rank of the read per independent channel without the addition of any learned parameters.

This necessitates a new architectural paradigm: **Route learned, collide fixed**. The routing of information (which tokens to attend to) must remain a learned, weighted operation, but the aggregation collision (how those tokens combine to form the output) should leverage fixed, deterministic, odd-parity structures. Replacing the learned, continuous bias with an object-side, deterministic collision mitigates the necessity for complex attention-sink denoisers and post-hoc frequency reparameterizations.

## 4. ARX Architectures, Carry-Bit Dynamics, and Cryptographic Hardness

To fully conceptualize the difference between arbitrary data access and cryptographic immutability, one must examine the routing of the arithmetic carry within Addition-Rotation-XOR (ARX) ciphers. Ciphers such as SPECK, Salsazo, ChaCha, and LEA rely exclusively on these three simple arithmetic operations, avoiding the use of traditional hardware-heavy S-boxes.

### 4.1 The Non-Linearity of Modular Addition

In an ARX cipher, the bitwise XOR and the bitwise cyclic rotations are entirely linear operations over  $GF(2)$ . They provide rapid, continuous diffusion across the block state. However, diffusion alone does not provide cryptographic hardness; a cipher constructed solely of rotations and XORs is perfectly bijective and trivial to solve via Gaussian elimination.

The sole non-linear component—and the exclusive source of the cipher's resistance to algebraic and differential attacks—is the modular addition operation ( $+$  modulo  $2^n$ ). During modular addition, the carry bit is not discarded; it is calculated and routed up the chain to the next most significant bit position.

### 4.2 Differential Cryptanalysis and the Long Trail Strategy

The security of ARX ciphers is evaluated through differential cryptanalysis, which seeks to trace how input differences ( $\Delta x$ ) propagate to output differences ( $\Delta y$ ) over multiple rounds. The differential probability of

addition modulo  $2^n$  is mathematically well-defined, and automated solvers utilize Satisfiability Modulo Theories (SMT) and Mixed Integer Linear Programming (MILP) to find the optimal differential trails.

A critical vulnerability in automated search methods has historically been the "independence assumption"—the premise that differential propagation across individual modular additions operates independently. This assumption fails catastrophically when a cryptographic trail includes Consecutive Modular Additions (CMAs). The routed carry bit creates a deterministic historical dependency. To accurately model this, cryptanalysts developed the Carry-Bit-Dependent Difference Distribution Table (CDDT), which proves that the state propagation relies entirely on the internal asymmetry generated by the carry.

As the number of rounds increases, the carry bit reaches near-maximal degree, causing the absolute correlations of linear approximations to decrease exponentially. This enforces the "Long Trail Strategy" in ARX design: maximizing the number of rounds ensures that the nonlinear propagation of the routed carry creates an inescapable, impenetrable wall of historical dependency.

#### 4.3 Carry-Route vs. Carry-Drop

The ARX architecture perfectly illustrates the "carry-route" paradigm. The carry bit represents the "exhaust" of one bit position being strictly forced to act as the "intake" of the next. In contrast, algorithms designed for arbitrary access and free traversal must employ a "carry-drop" paradigm. By comparing the cryptographic hardness of SHA/ARX to the spigot algorithms of number theory, the fundamental dichotomy of computational mechanics becomes clear: routing the carry creates an immutable historical ledger, while dropping the carry enables instantaneous phase navigation.

### 5. The Bailey-Borwein-Plouffe Formula as a Nulled Phase Comparator

The Bailey–Borwein–Plouffe (BBP) formula, discovered in 1995 by Simon Plouffe using Helaman Ferguson's PSQ integer relation algorithm, is widely celebrated as a spigot algorithm that computes the  $n$ -th hexadecimal digit of  $\pi$  without requiring the computation of any preceding digits.

Standard mathematical literature treats BBP primarily as an efficiency breakthrough in digit extraction and a tool for proving the base- $b$  normality of fundamental constants. However, an operational deconstruction of the formula reveals that it does not generate  $\pi$  through sequential construction; rather, it functions as a weightless, highly calibrated physical instrument—a nulled phase comparator measuring a pre-existing invariant structure.

#### 5.1 The Mechanical Decomposition of BBP

The BBP formula for  $\pi$  is formally defined as:

$$\pi = \sum_{k=0}^{\infty} \frac{1}{16^k} \left( \frac{4}{8k+1} - \frac{2}{8k+4} - \frac{1}{8k+5} - \frac{1}{8k+6} \right)$$

When analyzed through physical architecture rather than symbolic algebra, the formula disassembles into distinct structural components:

1. **The Drive (Geometric Taper):** The  $16^{-k}$  term represents an exponential geometric decay. Mechanically, this functions as a continuous logarithmic taper (a cone) or a tetherball winding around a pole, where the radius shortens by a factor of 16 with each recursive wrap. It serves as an attenuation cascade per stage.
2. **The Modular Rings (Phase Generators):** To extract the  $n$ -th digit, the formula computes  $16^{n-k} \bmod (8k + r)$  utilizing modular exponentiation. Each denominator defines a finite phase orbit, acting as a discrete gear or wheel circumference.
3. **The Taps (Phase Pickoffs):** The denominators  $8k + 1, 8k + 4, 8k + 5, 8k + 6$  define four fixed extraction phases within a period-8 cycle. These represent fractional angular offsets at exactly 12.5%, 50%, 62.5%, and 75% ( $45^\circ, 180^\circ, 225^\circ, 270^\circ$ ). These four channels do not operate as independent clocks; they are four simultaneous views of one continuous rotation.
4. **The Null Mixer (Differential Balance):** The coefficients  $(+4, -2, -1, -1)$  form a zero-sum bridge. Python execution confirms that the sum of these coefficients is exactly 0, while the sum of their tap positions equals the radix, 16.
5. **The Output Detent (Quantizer):** The continuous fractional phase result is multiplied by 16, and the integer part is skimmed off. This is physically equivalent to a 16-position Geneva mechanism or a spring-loaded pawl snapping into a discrete detent.

## 5.2 The Necessity of the Zero-Sum Condition

The coefficients  $+4, -2, -1, -1$  are frequently cited in literature merely as the output of the PSLQ algorithm. However, their summation to exactly zero is the critical mechanical property of the formula. In signal processing, an operator whose coefficients sum to zero acts as a precise DC blocker; it is structurally blind to uniform, constant backgrounds.

Because the instrument contributes zero net magnitude of its own, it functions as a weightless, untinted lens. A biased instrument with a non-zero sum would push the output toward a preferred internal attractor. By maintaining a perfect null equilibrium, the formula guarantees that any non-zero output is exclusively the pure differential relationship (the phase imbalance) of the underlying integer substrate.

## 5.3 Modular Exponentiation and the Dropped Carry

To achieve random access to the  $n$ -th digit, the BBP algorithm utilizes modular exponentiation, evaluated at the same loop level without nesting. The algorithm computes  $r = b^n \bmod m$  efficiently via binary exponentiation (square and multiply).

At every squaring step, the integer quotient is aggressively discarded. This is the exact inverse of the ARX cipher architecture. By continually dropping the carry, the BBP machine violently crushes the astronomical magnitude of the value into a constrained phase boundary. The mechanism becomes "weightless" because it refuses to accumulate the history of its own traversal.

## 5.4 The Power of Two Invariant and Base Independence

While the specific BBP formula for  $\pi$  operates strictly in base 16, PSLQ algorithms have discovered integer relations for other constants (e.g.,  $\ln 2$ ,  $\arctan(1/2)$ ,  $\zeta(3)$ ) operating on the same 8-stream basis using coefficient alphabets of  $\pm 8, \pm 4, \pm 2, \pm 1$ .

The true, unbreakable invariant across these formulations is the requirement that the base be a power of two ( $2^k$ ). In base 16, the fractional computation and the modular shift align cleanly without residue. The "fuel" (the positional shift mechanism) factors entirely out of the operation. Conversely, base 10 contains a prime factor of 5, which prevents a clean modular release. In base 10, the fuel contaminates the reading, the carry does not discard efficiently, and the exact spigot mechanism collapses. This enforces the architectural rule that an instrument must operate in a different basis from the substrate it measures to prevent self-referential loading.

## 6. Generalized Sampling and the Physicality of Noise

The architectural constraints observed in spatial topologies directly parallel the constraints governing signal processing and information theory. The Whittaker-Kotel'nikov-Shannon (WKS) sampling theorem famously dictates that a bandlimited signal must be sampled at a rate greater than twice its highest frequency ( $2f$ ) to allow for perfect reconstruction and to avoid aliasing.

### 6.1 Papoulis and Derivative Sampling

The conventional interpretation of the Nyquist limit treats the factor of two strictly as a temporal rate requirement. However, A. Papoulis' Generalized Sampling Expansion (1977) proved that any bandlimited function is uniquely determined by the responses of  $m$  linear systems, each sampled at exactly  $1/m$  the Nyquist rate.

A highly practical application of this is derivative sampling. A signal can be perfectly reconstructed using periodically nonuniform sampling (PNS) by acquiring both the sample of the function  $f(x)$  and the sample of its first derivative  $f'(x)$  simultaneously at exactly half the Nyquist rate.

This theoretical framework completely recontextualizes the "arity of the read" established in Section 2. The factor of two in the Nyquist theorem is not merely a count of consecutive temporal snapshots; it represents the two fundamental degrees of freedom inherent to a wave: amplitude (the value edge) and phase/slope (the change edge). A discrete operator that captures both edges simultaneously at one instant resolves the wave precisely, proving that temporal oversampling is simply an indirect mechanism for acquiring the relational degree of freedom.

### 6.2 Deterministic Aliasing and Structured Noise

When a structured signal violates the Nyquist bound relative to the sampling interval, phase-wrapping collisions occur, leading to positional aliasing. In classic signal processing, this aliasing is often categorized heuristically as degraded noise or irreversible entropy.

However, undersampling does not destroy the object data; it merely obscures the fold index. The folded frequencies map deterministically to exact integer grid points within the shift-invariant space. The resulting alias is highly autocorrelated and pixelated along a discrete lattice. Because the original signal was

computationally generated rather than randomly assigned, the pre-image set is highly constrained. Therefore, what is typically discarded as "thermal noise" or "random heat" is actually the original signal folded deterministically into the lower Nyquist band. Recovering the high-frequency structure from the low-frequency alias does not require massive temporal oversampling; it requires an orthogonal phase measurement (a rotational spread) to triangulate the implicit address that survived the fold.

## 7. The Automaton of Decimal Rounding

The extraction of computational mechanics extends down to the most foundational arithmetic interactions. The process of decimal rounding is traditionally treated as a static numerical approximation or a mathematically convenient truncation. However, when the fractional tail of a floating-point number is evaluated dynamically from right-to-left, the rounding mechanism reveals itself as a Turing-complete, programmable cellular automaton.

### 7.1 The Asymmetric 5-1-4 State Partition

In base-10 rounding, the ten available digits partition perfectly into a 5-1-4 asymmetrical logic engine. This asymmetry dictates how the recursive arithmetic carry propagates leftward across the string:

1. **0, 1, 2, 3, 4 (HOLD):** These digits act as absolute logic barriers. They absorb the recursive momentum, terminate the carry, and bring the cascade to a stable halt.
2. **6, 7, 8 (ROLL):** These digits represent the recursive delta. Upon absorbing a carry, they are forced to transfer a +1 carry to the left, acting as a standard propagation wave.
3. **9 (FREE CASCADE):** The digit 9 acts as a frictionless superconductor. Because it rolls over to 0 and passes the +1 carry leftward with zero mathematical residue, a continuous string of 9s serves as an unlimited-length transmission wire.
4. **5 (BRANCH):** The digit 5 sits precisely on the fault line of the asymmetry. It is a contextual logic gate that cannot resolve independently; it requires an external policy (e.g., ROUND\_HALF\_UP, ROUND\_HALF\_EVEN) to determine whether it collapses downward into a HOLD (4) or carries upward to trigger a ROLL (6).

### 7.2 Executable Logic in Fractional Space

This partitioning allows fractional decimal tails to function as runnable instruction sets. To empirically validate this, a Python simulation modeling right-to-left iterative rounding was executed on specific decimal configurations.



| Initial<br>Decimal<br>String | Iteration<br>1 | Iteration<br>2 | Iteration<br>3 | Iteration<br>4 | Iteration<br>5 | Final<br>Output |
|------------------------------|----------------|----------------|----------------|----------------|----------------|-----------------|
| 0.3444445                    | 0.344445       | 0.34445        | 0.3445         | 0.345          | 0.35           | 0.4             |
| 0.3444444                    | 0.344444       | 0.34444        | 0.3444         | 0.344          | 0.34           | 0.3             |
| 0.1999999                    | 0.200000       | 0.20000        | 0.2000         | 0.200          | 0.20           | 0.2             |

The execution of 0.3444445 demonstrates a delayed, sequential chain reaction. The single terminal BRANCH bit (5) resolves upward, sending a carry to the left. This carry strikes the adjacent 4 (a HOLD state), converting it to a 5. In the subsequent clock cycle, this new 5 resolves upward, striking the next 4. The wave unzips an entire static structure of holds, flipping them iteratively until the leading digit is pushed from 3 to 4. Conversely, changing the terminal digit to a 4 results in immediate termination, leaving the leading digit untouched.

This proves that continuous fractional rounding is a deterministic, object-side collision. Values are operators, and by pre-loading specific combinations of HOLDs, ROLLs, and BRANCHes, one can encode software directly into the topological shape of a fraction.

## 8. Conclusion

A mechanical analysis of computational algorithms reveals that structural topology, rather than symbolic data storage, dictates system behavior. Across cryptography, neural networks, integer relation mathematics, and signal processing, a unified set of architectural constraints governs the generation of valid outputs.

1. **Parity Dictates Preservation:** Even-arity operations inherently function as destructive derivatives, annihilating the fundamental carrier state. Odd-arity operations (such as a 3-copy rotational spread) preserve both relational difference and the baseline state, eliminating the need for external positional encoders in complex neural aggregations.
2. **Instruments vs. Generators:** The BBP formula does not sequentially compute  $\pi$ ; it operates as a nulled, 4-phase geometric comparator that measures pre-existing invariant cycles. Its ability to provide arbitrary digit access stems from continuously dropping the arithmetic carry, isolating the operation from historical dependencies.
3. **Hardness via Carry Routing:** In direct opposition to BBP, ARX cryptographic ciphers achieve immense computational hardness solely by routing the modular carry. This ensures that every bit is inextricably linked to the historical sequence of collisions, creating an impenetrable non-linear wall.

4. **Sampling and Automata:** Generalized sampling theorems confirm that temporal oversampling is merely a proxy for establishing the dual degrees of freedom inherent in a signal. Similarly, seemingly arbitrary mathematical conventions, such as decimal rounding, possess precise physical analogies to Turing-complete automata when executed recursively.

By analyzing computation as a series of forced geometric collisions, bounded constraints, and parity balances, the architecture of processing shifts from the manipulation of stored nouns to the continuous, deterministic resolution of shapes under stress.

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